

KUSH VASHISTH

+1 (213) 574-9863 | vashisth@usc.edu | Los Angeles, CA, USA | linkedin.com/in/kush-vashisth-52685117a |
github.com/kushvash | www.kushvashisth.com/

EDUCATION

University of Southern California
Master's, Computer Science

Jan 2024 - Dec 2025
GPA: 3.78

Gujarat Technological University
Bachelor's, Computer Science

Jul 2019 - Jun 2023
GPA: 3.65

SKILLS

- **Languages:** Python, JavaScript, TypeScript, C/C++, SQL
- **Backend & Infra:** FastAPI, Django, Node.js, REST APIs, AsyncIO, Redis, AWS, Docker, CI/CD, Linux/Unix
- **Frontend:** React, Next.js
- **Data & ML:** Pytorch, Transformers, NumPy, Pandas, Elasticsearch
- **Systems / Patterns:** Distributed workers, Task queues, Background jobs, Agentic workflow design

PROFESSIONAL EXPERIENCE

Sure Attend

Software Engineer Intern

Los Angeles, CA, USA

Feb 2025 - Aug 2025

- Developed Django APIs integrating Zoom, Calendly, and PayPal to automate meeting tracking and streamline backend workflows
- Deployed an ML model predicting user attendance using historical calendar and behavioral signals to improve scheduling decisions
- Raised test coverage by 40% with Pytest, reducing production bugs and improving long-term regression stability

University of Southern California

Teaching Assistant | Advanced Mobile Devices & Games (Academic Role)

Los Angeles, CA, USA

Jan 2025 - Aug 2025

- Delivered C# and Unity lectures to 120+ students in a graduate-level game development course, supporting design and implementation
- Conducted weekly code reviews to improve gameplay mechanics, performance, and overall project structure
- Evaluated 15+ student projects across development phases from Alpha to Gold, providing technical feedback and scoring

MBA & Beyond

Software Engineer Intern

Singapore

Feb 2022 - Jun 2022

- Built a workflow automation app in SailsJS, reducing manual email response time by 30% and improving operational efficiency
- Refactored Squarespace site components and added analytics-based user tracking, improving overall engagement by ~20%
- Collaborated with design and content teams to enhance UI consistency across mobile and desktop experiences

PROJECTS & OUTSIDE EXPERIENCE

Agentic Workflow Orchestrator – Local AI Automation System

- Built a fullstack agentic workflow engine (FastAPI, React, Redis) with Python worker processes for async multi-step execution
- Integrated Gemma2-2B (via Ollama) converting natural language into validated JSON workflows with schema enforcement
- Implemented a distributed execution layer with parallel workers, dependency-aware chaining, and real-time task progress
- Added an extensible tool registry and HTML/text preprocessing pipeline to support robust and adaptable automation modules

MediHear – Real-Time Doctor-Patient Teleconsultation Platform

- Developed with Next.js, WebRTC, Socket.IO, and Firebase authentication to support secure remote consultations
- Implemented real-time speech transcription using the Web Speech API and a Flask NLP service for live symptom extraction
- Achieved ~300 ms end-to-end latency for video and transcription in local tests, enabling natural conversational flow

PII Detection and Masking

- Built an end-to-end PII detection pipeline using PyTorch and Transformers to identify 94 PII categories across large corpora
- Fine-tuned DeBERTaV3, achieving 99.56% F1 with a 27% reduction in false positives, improving precision in sensitive-data tasks
- Integrated Elasticsearch to support scalable indexing and sub-second querying of millions of anonymized documents